Monday 12/10/2026

DATE / /

Case Study : 2

Securing the University Examination System

A university uses an OS-based examination system to store students records, examination marks, faculty details, & questions papers. The system has 3 main user groups: Students, faculaty, & Administrators. Students should be able to view only their own marks and download permitted learning materials. Faculaty members need to view students records and enter or modify marks. Administrators manage user accounts, permission, & system configurations.

The university has assigned permissions through roles, but an audit review identifiers the following events:

- At 10:00, the account student07 has 3 failed login attempts.
- At 10:03, Students07 sucessfully logs in

DATE / /

- At 10:04, Students 07 attempts to open Exam marks.xlsx, which contains marks for an entire class.
- At 10:05, the system records an attempted modification of Exam Marks.xlsx by Student 07.
- At 10:15, Faculty 02 modifies ~~record~~ marks in Exam Marks.xlsx ~~and s~~ and saves the file succesfully.
- At 10:30, an administrator discoveres that a former faculty ~~mem~~ member's account is still active.
- The audit logs is stored on the same server, and users with broad permissions may be able to modify or delete it.

The university wants to protect confidentiality, integrity, and availability, prevent unauthorized access; maintain accountability; and investigate ~~susp~~ Suspicious activity. You are asked to recommend appropriate OS access-control, security-administration, and auditing measures

DATE / /

* Questions

Q.1) Identify the main security problems in the case study.
[List at least four risks or weakness incidented by the scenario]

Q.2) Explain the difference between authentication and authorization using student ID as an example.
[Distinguish identify verification from permission checking]

Q.3) Recommend an access-control policy for students, faculty, & administrators. Justify your choice.
[Consider how permissions should be assigned & managed]

Q.4) Prepare a suitable Access Control list [ACL] for Exammarks.xlsx.
[Specify appropriate Read, write, & Delete permissions for each user group]

Q.5) Explain how the principle of least privileage should be applied in this system.
[State what each groups needs to be do & which unnecessary permission should be removed]

DATE / /

Q.6) Interpret the audit events from 10:00 to 10:05 what should the security teams do next?
[Explain why the sequence is suspicious without assuming that it proves an attack].

Q.7) Identify at least five events that should be recorded in the audit logs.
[include login, file, account, permission, and/or administrative events.].

Q.8) Explain why the audit logs must be protected and recommend two ways to protect them.
[Consider unauthorized modification, detetion, & investigation].

Q.9) What action should we be taken regarding the former faculty members active account? Explain
[Relate your answer to account management & access removal]

Q.10) Suggest a practical incident-response & corrective-action plan for the university
[Include investigation; access review, policy changes, & & follow up monitoring]

Answer

1) Four risk are:
i) At 10:00 student 07 get 3 attempts to login, this problem indicate unauthorized login.

ii) Immediately after, the user attempted to open Exam marks.xsls.

iii) Edit logs which student 07 edit & modify.

iv) An ex faculty account is still active on system. which should be removed.

2) Authentication :- Student 07 sign in at 10:03 with it's username & pass, the system verifies ~~their~~ it's identity & grant entry.

Authorization:-
The system enforces access rules ~~Ev~~ even though student 07 is verified, authorization restricts them to change ~~thei~~ their own records.

DATE / /

3) • The university should implement Role Based Access control.
• Students only have visibility to their ~~past~~ passed records.

• Faculty get access exclusively for the classes they teach.

• Admin maintains user lifecycle & system settings rather than day to day grade editing.

4)

Task	Student	Faculty	Admin
View own marks	Allow	Allow	Allow ~~All if required~~
View all marks	Deny	Allow	Allow
Edit Marks	Deny	Allow	Allow
Delete records	Deny	Deny	Allow
Manage user	Deny	Deny	Allow
View logs	Deny	Deny	Allow.

5) Students should only have access their own academic records.

Faculty have access to their assigned course.

Admin receive only the permission required to manage acounts

DATE / /

6) The 3 failed login attempt by student 07 at 10:00 then logs in at 10:03, & attempts to modify Exam marks.xlsx at at 10:04 - 10:05.

The Security Team Should:
i) Review audit logs.
ii) Verify if the logins were legit
iii) Checks whether records are not changed.
iv) Temporarily restrict the account
v) Do audit log & investigate Preserve.

7) i) login & logout events
ii) file access events
iii) Account Management events.
iv) Admin activity
v) Security alerts.

8) It must be to protected because they provide evidence of user activity & & security incidents.

Two ways to protect.

i) Restrict access, only security personnel can access.
ii) Store logs on seperate, centralised Server.

DATE / /

9) i) Disable the account.
ii) Remove access to university portal.
iii) Review recent login history
iv) Preserve audit logs.
v) Update account Management.

10) i) Identification :- Examine audit logs & identify it.
ii) Containment :- Restrict compromise account
iii) Investigation :- Preseve audit logs.
iv) Recovery :- Restore affected records
v) Policy changes :- Role based access control.
vi) follow up monitoring :- monitor audit logs.
vii) Documentation :- Inform higher authority.